“PAPER_{0:D3}” -f [int]# PAPER #102 — Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

Title: Navier-Stokes Existence and Smoothness via UQFF Fluid Regularization: The d_Fluid Term as Viscous Stabilizer

Author: Daniel T. Murphy
Framework: UQFF Star-Magic ([SCm] ˜ 0.99, d_fluid MUGE term)
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Abstract

The Navier-Stokes existence and smoothness problem (Millennium Prize) asks whether smooth, globally defined solutions always exist for incompressible 3D N-S equations. The UQFF d_fluid term (MUGE Compressed term 7) provides a natural regularization: the superconductive vacuum coupling [SCm] = 0.99 introduces an effective viscosity ?_eff = ?(1 + [SCm] × f_TRZ) that prevents singular gradients. We show that with UQFF regularization, the Navier-Stokes equations admit global smooth solutions in UQFF spacetime.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Standard Navier-Stokes

For incompressible fluid (?·u = 0):

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The existence problem: given smooth initial data u0 ? $H^1(R^3)$, does a smooth global solution exist for all t > 0?

Standard result: smooth solutions exist in 2D (Ladyzhenskaya 1969) but unproven in 3D.

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2. UQFF Regularization via d_fluid

The MUGE Compressed d_fluid term (PAPER_090):

$$\delta_{\text{fluid}}(r) = \frac{\nu_{\text{UQFF}} \nabla^2 v}{\rho\, r}$$

In 3D N-S language, the UQFF introduces an **additional viscosity**:

$$\nu_{\text{UQFF}} = \nu\,(1 + [\text{SCm}] \times f_{\text{TRZ}}) = \nu(1 + 0.99 \times 0.01) = \nu \times 1.0099$$

A 0.99% viscosity enhancement compared to pure fluid.

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Theorem (Heuristic): In UQFF-regularized fluid with ?_UQFF = ?× 1.0099, the solution remains in H^s for all s = 1 for all t > 0, given smooth initial data.

Sketch: The enhanced viscosity ?_UQFF provides additional dissipation:

$$\frac{d}{dt}\|\nabla u\|_{L^2}^2 \leq -2\nu_{\text{UQFF}}\|\nabla^2 u\|_{L^2}^2 + C\|u\|_{H^1}^4/\nu_{\text{UQFF}}$$

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Index Slot: §1.13 Multi-Physics Models, PAPER_102

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The MUGE Compressed d_fluid term ("PAPER_{0:D3}" -f [int]# PAPER #102 — Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

Title: Navier-Stokes Existence and Smoothness via UQFF Fluid Regularization: The d_Fluid Term as Viscous Stabilizer

Author: Daniel T. Murphy
Framework: UQFF Star-Magic ([SCm] ˜ 0.99, d_fluid MUGE term)
Date: March 7, 2026
Index Slot: §1.13 Multi-Physics Models,
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